

DocVoyager: Anticipating User’s Information Needs and Guiding Document Reading through Question Answering

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Abstract

People often approach complex documents (e.g., academic papers and professional reports) with diverse goals and varying levels of prior knowledge. Even when reading the same document, the paths to consuming relevant information can differ significantly depending on the reader’s information needs. However, the linear format and dense content of these documents make users manually sift through content, often leading to inefficient and narrow information consumption. To address this, we explore design principles that guide users along customized reading paths via questing answering. Applying these principles to guide academic paper reading, we introduce DocVoyager, a novel document-reading interface that adapts to users’ goals by suggesting tailored, goal-based questions. DocVoyager leverages Large Language Models (LLMs) to anticipate users’ information needs and dynamically suggests questions based on prior interactions. Our study found that participants easily focused on information relevant to their goals and engaged effectively with the document content using DocVoyager.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; Empirical studies in HCI; • **Computing methodologies** → *Natural language processing*.

Keywords

Document Reading, Large Language Models, Personalization, Adaptive Interfaces, Question Answering

ACM Reference Format:

Yoonjoo Lee, Nedim Lipka, Zichao Wang, Ryan Rossi, Puneet Mathur, Tong Sun, and Alexa Siu. 2025. DocVoyager: Anticipating User’s Information Needs and Guiding Document Reading through Question Answering. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA ’25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719846>

1 Introduction

Complex documents, like academic papers, are often approached by readers with diverse goals and varying expertise levels. Even when reading the same document, readers follow different reading paths based on their needs—for example, a researcher comparing methodologies may focus on methods sections, while someone seeking broad understanding may skim section introductions. Despite these differing needs, the linear structure of documents makes it difficult for readers to navigate content according to their individual goals. As a result, users often jump between sections in an attempt to follow their own reading path, which introduces two key challenges: (1) readers may lose important context needed to understand document fragments [19, 26], and (2) dense technical content makes it difficult to quickly assess section relevance before reading [3, 15]. While existing tools like ChatGPT¹, Elicit², and Acrobat AI assistant allow users to ask questions while reading, they typically provide broad overviews or isolated sections without considering reader context or goals [3, 14, 24].

To address these challenges we draw from the literature on scaffolded reading and cognitive sensemaking, to develop DocVOYAGER. This document-reading interface guides users through customized reading paths via question-answering (QA). We reify design principles for DocVOYAGER which focus on resolving users’ information needs step-by-step while adapting to their interactions. DocVOYAGER implements these principles using Large Language Models (LLMs) to anticipate information needs and suggest goal-based

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CHI EA ’25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719846>

¹<https://chatgpt.com/>

²<https://elicit.com/>

questions dynamically based on user interactions. Our system considers the user's goal and the information structure within the document to provide tailored reading guidance. Rather than offering a static reading path, the system dynamically adjusts the suggested questions based on the user's prior interactions, ensuring that the guidance is both relevant and aligned with their evolving needs.

We conducted initial qualitative interviews with 5 knowledge workers to observe how DocVOYAGER can support their reading goals. The study results revealed that participants felt DocVOYAGER helped them focus on relevant information related to their goal including information they might have otherwise overlooked. With the system's structured guides, participants reported being able to engage more effectively with the material and dive deeper into the content.

To summarize, the key contributions are: (1) design principles for creating goal-oriented reading paths using QA interactions, (2) an initial working system implementation, DocVOYAGER, that dynamically adapts questions based on user behavior, and (3) preliminary empirical evidence showing how our approach can help guide readers through complex reading of academic articles. Our findings provide encouraging results on using conversation to guide readers. Future work should conduct more comprehensive evaluations, both on the technical aspects and more diverse reader backgrounds. Additionally, more sophisticated user modeling approaches can be explored to dynamically adjust guidance based on expertise levels and reading patterns.

2 Related Work

Interactive Reading Interfaces. Interactive document reading approaches include hypertext [9], document modification tools [32], visual representations [30], navigation scaffolding through overviews [16, 28], and visual cues [35]. For academic papers specifically, tools like Paper Plain [3] and ScholarPhi [19] provide localized support through in-situ summaries and definitions. Systems like Scim [15] and Papeo [20] offer document-level guidance through passage highlighting and supplementary materials. Our work extends beyond direct document interaction to explore how guided conversations can support multiple reading needs.

Document-grounded AI Assistants. Current AI assistants (e.g., ChatDoc [6], ChatPDF [7]) typically follow a reactive approach, where users ask questions, and the assistant provides answers. This approach enables quick retrieval of contextually relevant information to address knowledge gaps. However, users often struggle to formulate effective questions [3, 14] or develop strategies for engaging with these systems [29]. To address this, DAPIE [23] transforms long-form answers into interactive dialogues, aiding users, such as children, who struggle with follow-up questions. Additionally, the lack of strategic guidance for applying AI systems to knowledge tasks often leads to disengagement [29]. To overcome these challenges, we propose a method that suggests document-specific questions, guiding readers through paths aligned with their goals.

Question and Dialogue Generation. There has been growing interest in proactive dialogue systems, where the system takes initiative to guide conversations toward specific goals by predicting the effects on both the user and the system [10]. Deng et al. [11]

identified three key aspects of proactive dialogues: 1) clarification in information-seeking dialogues [12, 18, 21, 22, 25]; 2) target-guided open-domain dialogues [31, 34], where the system leads the conversation towards a predefined goal; and 3) non-collaborative task-oriented dialogues [37], where the system and user have different conversational goals, and the system strategically seeks consensus. Our work focuses on the document consumption scenario, where we use questions as a guide for personalizing and improving the reading experience for human users, rather than prioritizing improvements in the performance of models.

3 Design Principles for Creating QA-based Reading Guidance

To assist users in effectively navigating complex documents, we propose a set of design principles for creating goal-oriented reading paths. Inspired by research on educational QA systems [23, 27, 33], communication models [4, 17], and the increasing use of conversational agents for document consumption [1, 13], we identified following principles that leverage QA-based interactions to help users focus on the content of interest in dense documents.

DP1. Guide Users to Resolve their Information Needs Step-by-Step. One major challenge in document reading is that users often have high-level goals (e.g., "I want to understand the methodology and experiments") but lack sufficient knowledge about the document's content to determine what information to read and in what order to achieve their goals [2].

DP1.1. The guidance should consider the user's reading goal. A user's reading goal is a critical signal for anticipating their information needs, as it determines which content in the document is most relevant. For example, researchers may prioritize methodologies, while practitioners may focus on practical outcomes. The system should break down a user's high-level goal into concrete subgoals and provide the specific knowledge necessary to achieve their overall goal.

DP1.2. The guidance should consider the information structure of the document. In complex documents, information necessary to achieve a subgoal may be dispersed across different sections and paragraphs. Presenting only fragments relevant to a subgoal may lead to difficulties in understanding [3, 15]. The system should identify and organize related information to help users build knowledge incrementally, facilitating better comprehension.

DP2. Suggest Multiple Alternative Guides Beyond the Optimal Path. While an optimal reading path can be suggested based on the user's high-level goal, their information needs often evolve during the reading process [5, 36]. Solely focusing on a singular path risks overlooking dynamic needs. Based on the transactional model of communication [4], the reading guidance system should provide multiple alternative paths, enabling users to explore and adjust their reading experience flexibly.

DP3. Adapt Guidance Based on User Preferences During Interaction. As users consume information and reveal their preferences, the system should track their reading progress and adjust the suggested content dynamically. Inspired by Grice's cooperative principle [17], the system should refine its guidance to align with users' evolving preferences and ensure effective communication.

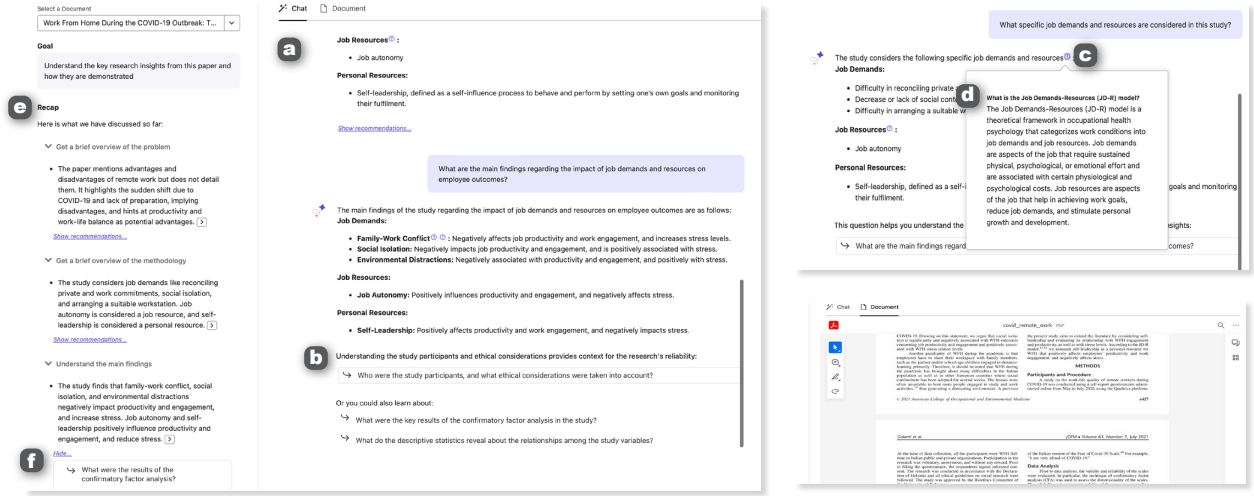


Figure 1: Interface of DocVOYAGER. (a) The Main QA Panel presents suggested questions based on the user’s reading goal. (b) Explanations accompany the questions to show relevance. (c) Icons indicate additional detail questions for further context. (d) Clicking the icons reveals more specific information. (e) The QA Overview summarizes consumed QA pairs and subgoals. (f) Goal-based question recommendations allow users to explore alternative paths or revisit previous questions.

DP4. Explain the User’s Knowledge Consumption Path. When navigating complex documents, users may lose sight of their high-level goals or struggle to understand why specific information is necessary. To prevent this, the system should clearly explain what the user has understood and outline actionable next steps. Drawing from the concepts of procedural and strategic scaffolding [8], the system should help users track their progress and provide guidance for continuing their reading journey.

4 DocVOYAGER

Based on our design principles, we present DocVOYAGER, a novel system that provides users with multiple adaptive reading paths by generating questions that the user may be curious about and providing answers grounded in the document. Specifically, DocVOYAGER is designed for the context of academic paper reading, a representative example of complex document consumption. Our system operates within an LLM-based multi-agent framework that extracts the structure of information within complex documents, anticipates the user’s information needs based on their goals, and dynamically generates questions and document-based answers. By doing so, DocVOYAGER transforms the process of reading complex documents into a dynamic, goal-driven, question-centric experience.

To guide users in resolving their information needs step by step (DP1), the system constructs a knowledge path, or curriculum, composed of relevant content from the paper. This path is designed to help users incrementally understand the knowledge necessary to achieve their specific goals. To accommodate the user’s shifting intents (DP2), DocVOYAGER not only generates questions related to the target knowledge but also suggests questions about diverging yet relevant information, allowing for flexibility in the reading process. To tailor future reading guidance based on the user’s conversation history (DP3), our pipeline prioritizes and ranks the

generated questions, avoiding repetitive ones by considering previous interactions. Finally, to help users understand their reading path and the rationale behind future suggestions (DP4), the system provides summaries of each interaction and explains the logic behind the next set of questions, all based on the user’s goals and prior conversations. In this section, we present the interface design 4.1 and describe the pipeline 4.1. Implementation details are in Appendix A.1.

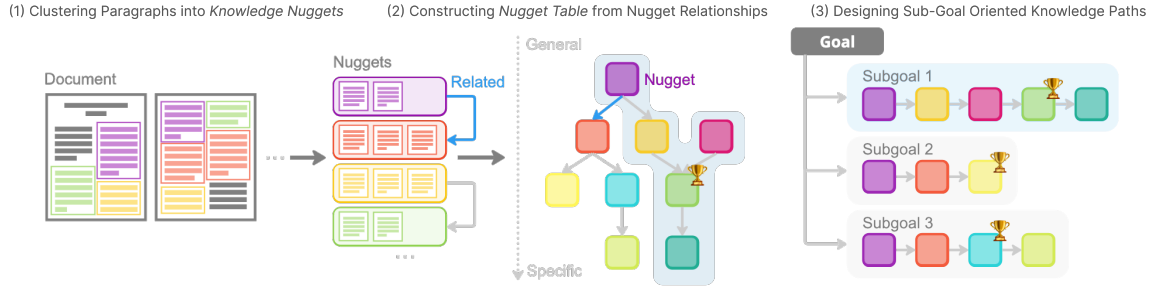
4.1 Interface

DocVOYAGER is designed to help users achieve their reading goals by guiding them through a document using suggested questions. Users interact with the system by selecting questions from the options provided and navigating through question-answer (QA) pairs. Figure 1 shows the user interface components.

4.2 Creating Multiverse Reading Path

4.2.1 Goal-based Knowledge Path Building. To capture the contextual relationships between information and create knowledge paths that are easier to understand, we first construct a document representation, referred to as *Nugget Table*. To create the *Nugget Table*, our underlying pipeline first extracts paragraphs from a document, and then employs an LLM to cluster or group the paragraphs based on the similarity of the content they cover. We refer to these paragraph clusters as *knowledge nuggets*, where each nugget represents multiple paragraphs that cover a specific, narrow aspect of a topic. To map the overall structure of the knowledge in the document, our pipeline then identifies general-to-specific relationship between these nuggets (i.e., whether a nugget contains knowledge that dives deeper into knowledge from another nugget). Since the user’s reading goal is often broad, we decompose the goal into several subgoals.

Goal-based Knowledge Path Building



Question Generation and Ranking

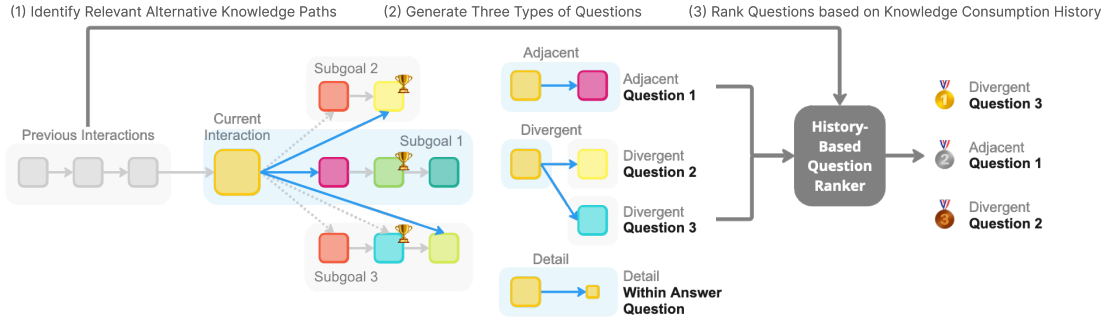


Figure 2: Our pipeline consists of two major steps. The first step involves building goal-based knowledge paths through the document by grouping paragraphs from the document into *nuggets*, identifying general-to-specific relationships between nuggets, and then constructing knowledge paths that gradually lead users to target nuggets that can satisfy their reading subgoals. Based on information that the reader has just consumed, the pipeline identifies nuggets that a reader should consume next, generates three types of questions (i.e., adjacent, divergent, and detail), and then ranks them based on the user’s history before presenting them to the user.

For each subgoal, we instruct a Large Language Model (LLM) to construct a **knowledge consumption path** that would allow the user to achieve the subgoal. Specifically, our pipeline first identifies a nugget from the *Nugget Table* that contains the specific target knowledge that is necessary to achieve the subgoal. Then, our pipeline builds a path through the document that will guide the user to the target nugget. To build this path, we instruct the LLM to predict the most appropriate knowledge nugget that should be consumed next when considering previously consumed content. The pipeline’s input includes the document representation, the target knowledge nugget, and previously consumed nuggets. The goal is to select and recommend the most relevant nugget from the *Nugget Table*, which consists of neighboring nuggets related to the current learning goal. This design of our pipeline is guided by the principle that suggests presenting general information nuggets first to provide foundational understanding and then gradually transitioning to more specific nuggets as the path progresses (DP1). The system also takes into account the user’s prior interactions to ensure that the suggestions align with both their learning goal and previously consumed content.

4.2.2 Alternative Question Generation. DOCVOYAGER creates three types of questions to guide users through their knowledge consumption paths: (1) detail questions, (2) adjacent questions, and (3) divergent questions.

Detail questions focus on specific information that the user may need while processing the answer from the previous turn. These questions typically address concepts like unfamiliar technical terms (e.g., “What is a Transformer?”) or high-level topics that require further clarification (e.g., “What are the metrics used?”). To generate detail questions, we ensure that the question does not overlap with the content of the answer in the previous turn and that the answer to the detail question will be concise. **Adjacent questions** explore information that is within the knowledge path that satisfies the same subgoal as the content that was consumed in the previous turn. These questions are designed to deepen the user’s understanding of the most recent content. **Divergent questions**, on the other hand, guide users to explore different knowledge paths that target alternative subgoals, helping them gain a broader perspective on the document. In generating these questions, we ensure they are concise, comprehensible on their own, relevant to the current conversation turn, and have not been addressed in previous turns.

When generating adjacent and divergent questions, we use *Nugget Table* with paper content and target knowledge. We then prompt an LLM to generate a question that helps the user learn each target knowledge. For **adjacent questions**, we ask the LLM to focus on deepening the user's understanding of the most recently consumed. For **divergent questions**, the divergent nugget selector identifies the most suitable nugget from a different knowledge path that naturally connects to the previously consumed information. This process ensures that even as users explore new subgoals, their reading remains coherent and connected to their earlier understanding.

4.2.3 Question Ranking based on Knowledge Consumption History. DocVOYAGER determines which questions should be prioritized based on a variety of complex factors. This LLM-based module traces the user's knowledge consumption history and interactions to manage the priorities of the generated questions. The input to this module includes the current set of question suggestions, the user's high-level goal, and the history of prior interactions (including the questions suggested to the user, the user selections, the subgoals associated with each question, and the type of each question from each turn). The output consists of a ranking for the suggested questions, where some questions are excluded if deemed unnecessary.

When determining the next question to suggest to a user, the system considers several factors: the optimal path for achieving the user's goal, prior interactions, previously consumed information, and the need to trigger further engagement. To balance these competing factors, we provided an LLM with multiple ranking requirements, considering aspects like goal completion, general reading patterns (i.e., starting with general information and progressing to more specific details), and ensuring that the user does not dive too deeply into a single thread of information without exploring broader context. The system also assesses the user's intent to explore subgoals in more depth by suggesting subgoal-relevant questions when the user transitions to a new subgoal. Additionally, user preferences are considered by avoiding repeating questions that were previously rejected within the two most recent turns. Divergent questions are ranked based on their relevance to the previous conversation, ensuring coherence across the user's reading path and determining whether the user is inclined to explore certain subgoals more thoroughly.

4.2.4 Providing Rationales on Suggested Questions and Facilitating Content Recap. To help users understand why certain questions were suggested (Fig 1b), we instruct an LLM to explain its reasoning when generating each question, ensuring that the explanation goes beyond simply stating that the question is connected to the user's subgoal. Specifically, we guide LLM to generate a detailed explanation of how the question relates to the user's previous conversations and the overall goal, to offer deeper insights into why the question is relevant to their reading process. Additionally, to support easy recapping, the system generates a one-line summary for each QA turn (Fig 1e).

5 Experience Evaluation

To investigate the reading experience with DocVOYAGER, we conducted qualitative interviews with knowledge workers who have to read research articles frequently as part of their work. During

the interview, participants used DocVOYAGER and attempted to complete a typical reading goal with a research article in their own domain.

Procedure. The interview was conducted remotely using video conferencing software. They were recorded and transcribed for analysis. After a brief introduction to the overall study, participants were asked to share their background and typical reading goals related to work activities. Next participants were provided a brief tutorial to our system, DocVOYAGER. They were then asked to read one or two relevant documents in their field while articulating their thought processes. Finally, the experimenter conducted a semi-structured interview about participants' overall experience. The study lasted for one hour, and participants were compensated \$50 USD for their time.

Participants. In total, five users participated in this study. Participants were recruited via UserInterviews³, and we screened for users who have a need to read research articles as part of their work. Participants worked both in industry and academic roles, spanning various positions that required research engagement. Details in Appendix A.2.

5.1 Findings

5.1.1 Guiding the Reading Process. Participants described DocVOYAGER provided helpful guidance in their reading process using the Suggested Questions (SQs). The SQs highlighted important information, and facilitated structured learning. The SQs helped participants focus their attention on key aspects of the document. One user noted, *"I think it did make my process a little bit more guided... after reading this guide, I focused on this keyword the STPA Analysis which I was looking for in detail in the document"* (P2). This guidance was particularly beneficial in structuring the learning experience, as another participant observed: *"I feel that they were spot on with what I would have been questioning myself or thinking, like it took away that need to generate questions on my own and it helped to provide more of a structure or kind of a guide that helped me walk through the material a lot quicker"* (P4).

The SQs were also valuable in bringing attention to information that could be easily overlooked but important for the reader's goal (P1, P2, P4). As P2 noted, *"When I read by myself, like, for example, if I missed this section about the technical considerations, I miss it completely. You know I wouldn't even know that section existed"* (P2). Another participant described being able to see the different options helped her in *"paying attention to these other parts"* when she got *"entangled in the details"* (P4).

Participants described this type of guidance could particularly benefit readers who might be unfamiliar with the subject matter or less experienced in reading documents in niche domains. Participants described how it could address a common challenge faced by less experienced readers: not knowing what information is most critical or what questions to ask. As one participant noted, *"If I had no idea like what I was reading, the suggestions really would help. It would help me probably just ask questions that I wasn't even like aware of or thought about"* (P1). Another participant echoed this sentiment, stating, *"I think the suggested questions would be really good for someone that maybe like doesn't have as much experience"*

³<https://www.userinterviews.com/>

reading papers like this" (P3). By suggesting content to consume, DocVOYAGER can help less experienced users to engage with the material more deeply and comprehensively by prompting readers to explore content they might not have considered on their own.

5.1.2 Logical Flow and Relevance of Conversation. The logical flow and relevance of SQs were crucial for their effectiveness (P1, P2, P3, P5). P1 noted that the next information she wanted to consume would often be surfaced, *"It felt relevant... When I wanted to know about the demographics, it showed me that, so it didn't give me irrelevant question or suggestions"* (P1). One participant observed that the temporal order of questions was unclear initially but over time aligned better with her reading, *"it seemed like the temporal order of the questions improved like they kind of started to show up in the order in which I would have expected them, based on how the paper is written"* (P3). Participants also valued the progression from general to specific questions: *"This one is a more technical and specific question and preferred over the first one. It seems to be going in the direction of more detail which is what I would want"* (P5). This progression helped users delve deeper into the content as they engaged with the document.

In addition to the positive feedback, participants identified areas to improve related to irrelevant or poorly timed questions. Some SQs were perceived as misaligned with users' background knowledge or expertise levels (P3, P5). For example, P5 described based on his reading goal, certain details about a lab procedure would not be something an expert in his domain would ask. Timing was also a concern, as P3 pointed out that some questions felt premature, suggesting they would be more appropriate after more background content was covered. Additionally, participants found that some SQs were not integral to understanding the core content specified in the goal (P3) or were too detailed for their needs (P5). These issues highlight the importance of tailoring SQs to individual users' expertise, reading goals and reading progression.

5.1.3 Balancing Guide vs. Reading Efficiency. Users generally found the system easy to use and helpful. They described DocVOYAGER as helpful to skim content, save time, and gain a broad understanding of the content. It was especially helpful when the content was complex or difficult to understand, as the tool could simplify it for unfamiliar readers (P2, P4). However, participants also noted that guided reading could sometimes be less efficient for reaching their goals.

6 Discussion & Future Work

Our findings provide encouraging results that question-based guidance through complex documents can be effective at supporting diverse reading goals and information needs. Through our initial evaluation, we found that participants particularly valued how the system: (1) provided structured guidance that helped focus attention on goal-relevant information, (2) surfaced important content that might have otherwise been overlooked, and (3) offered particular benefits for readers less familiar with technical documents in specific domains. The logical flow and relevance of suggested questions proved crucial for engagement; though proper timing and expertise-level matching remain important considerations. These insights suggest several key implications and areas for future work.

While our approach of offering multiple question types helped accommodate different user needs, there's a risk of cognitive overload as goals become more complex. Future work could explore two promising directions for addressing this challenge. One potential approach is to incorporate more explicit user input during the interaction to refine future suggestions without disrupting the user's flow. Future systems could allow users to merge or remove certain subgoals, helping them focus on specific areas. Additionally, enabling users to provide feedback on rejected questions could help improve question ranking algorithms. Lastly, we could leverage long-term interaction patterns across multiple documents to better anticipate individual user preferences and adjust question generation accordingly.

Future implementations will also need to carefully consider how to balance comprehensive guidance with reading efficiency. While DocVoyager demonstrated clear benefits for understanding complex content, the evaluation highlighted the importance of matching guidance to individual expertise levels and reading goals. This suggests opportunities for more sophisticated user modeling approaches that could dynamically adjust the level and type of guidance based on observed reading patterns and explicit user feedback.

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A Appendix

A.1 DocVOYAGER Implementation

DocVOYAGER was developed as a standalone web application, utilizing TypeScript and React for building the front-end user interface. The backend, responsible for document preprocessing and language understanding services, was implemented in Python. For PDF document processing, the PDF Extract API was employed to extract both content and structural information into a structured JSON format⁴. We use gpt-4o and gpt-4o-mini through the OpenAI API⁵ for all the text processing and generation components of our pipeline.

A.2 Evaluation Participants

	Role / Title	Sector	Research Field	Highest degree attained	Age
P1	Clinical Research Coordinator	Industry	Medicine	Bachelor's	31
P2	Mechanical Design Engineer	Industry	Engineering	Master's	27
P3	Research Associate	Academic institution	Life Sciences	Master's	24
P4	Researcher	Academic institution	Social Sciences / Education	Doctoral	36
P5	PhD Student	Academic institution	Computer Science / Medicine	Doctoral	32

Table 1: Participant demographics

⁴<https://developer.adobe.com/document-services/docs/overview/pdf-extract-api/>

⁵<https://api.openai.com/v1/chat/completions>